Renderoc 20

constructive solutions

Multi-purpose cementitious repair mortar

Uses

Renderoc 20 is ideal to be used in following typical applications in civil construction, industrial and buildings

- Repair of mechanically damaged concrete.
- Repair of concrete affected by reinforcement corrosion.
- General repair and restoration work on concrete.
- Filling tie-bar holes.

Advantages

- Ready to use, simply add water.
- For exterior and interior use.
- Excellent adhesion to concrete and reinforcing bars.
- Excellent impact strength.
- Low permeability helps minimize chloride intrusion.
- Protects reinforcement from chlorides and water ingress.

Description

Renderoc 20 is a high strength repair mortar containing blended cements, graded sands and chemical additives, which provide a mortar with the superior properties required for durable structural repair.

Renderoc 20 supplied as a ready to use blend of dry powders which requires only the site addition of clean water on site.

Typical Properties

Properties obtained in laboratory conditions at ~25°C

Wet Density	:	2300 Kg/m^3
Phase	:	Powder
Setting Time	:	Initial 45 min. Final 3 hours at 25°C
Compressive strength (ASTM C579) After 28 days	:	>35 N/mm^2
Tensile strength (ASTM C307) After 28 days	:	>10 N/mm^2
Flexural strength (ASTM C580) After 28 days	:	>9.0 N/mm^2

Instructions for use

Preparation of concrete surface

All substrates must be clean, sound and should be roughened sufficiently to provide a mechanical key between the original materials.

The damaged areas of concrete to be removed should be clearly identified. The perimeter of the area should be saw cut to a depth of 10 mm and the edges cut as neatly as possible keeping the sides square.

Clean the surface and remove any dust, unsound or contaminated material, plaster, oil, paint, grease, corrosion deposits or algae.

Feather edging is not permitted and a minimum thickness of 10 mm must be maintained over the whole repair area.

Preparation of reinforcing steel

Surfaces should be prepared using abrasive blast cleaning techniques or high pressure water blasting. Remove all rust and scale from the exposed reinforcing steel.

If reinforcement is corroded, ensure that the back of the steel has been exposed. Reinforcement should have all rust removed by the use of power tools, abrasive blasting (wet or dry) or wire brushing. Reinforcing steel should be exposed and cleaned around its whole circumference.

Where exposed reinforcement is contaminated with chloride or other material which may cause corrosion, the reinforcement shall be cleaned by low pressure water-blasting

For additional protection from future corrosion, coat the prepared reinforcing steel with Nitoprime Zincrich.

Bonding primer

On a well-prepared and roughened substrate the surface should be pre-wet with clean water.

It is recommended to use a bonding primer for concrete surface with using Nitobond SBR water dilution rate of 1 part of bonding agent to 4 parts to water.

The surface should not be allowed to dry before application of the concrete repair mortar. Renderoc 20 should be applied while primer is still wet or tacky. If the primer dries, re-prime the surface.

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Mixing

It is essential that quantity of more than one bag are mixed in a forced pan mixer. Individual bags may be mixed with a slow speed drill (apx.500 rpm) for 3 - 4 minutes until mortar is homogeneous and lump-free.

Always add powder to water and mix until homogeneous. The water must be clean, uncontaminated, and should be maximum of 4.25 litres per bag (25 kg).

Application

After mixing, Renderoc 20 should be compacted by trowel or gloved hand onto the primed substrate. Care should be taken to ensure full compaction around any steel reinforcement.

If sagging or slumping occurs, remove the mortar and re-apply at a reduced thickness.

Maximum thickness in a single layer should not exceed 30mm. Additional build-up can be achieved by the application of multiple layers. The surface of intermediate layers should be left rough or scratch finished and allowed to set. The substrate shall be re-primed before the application of the next layer.

Finish the completed repair, as required, taking care not to overwork the surface.

Curing

Renderoc 20 should be cured immediately after finishing with standard procedures and using Fosroc curing products.

Limitations

The maximum thickness in a single layer is 3 cm.

Renderoc 20 can be used when the ambient temperature is between 5 and 50°C.

High Temperature Working

At Temperatures above 35°C the grout packages should be stored in cool conditions and cold water (<20°C) should be used for mixing. Consult with Fosroc Technical department for further details.

Estimating

Supply and Yield

Renderoc 20 is supplied in 25 Kg plastic bags.

Rates: One bag (25 kg) of Renderoc 20 covers approx. 12.5 litres of mortar, this will cover 1m² at 12.5 mm thickness.

Storage

Renderoc 20 must be stored in a cool, dry store away from sources of heat in the original unopened packaging.

Precautions

Health and safety

For further information consult the Material Safety Data Sheet available for this product.

Fire

Renderoc 20 is non-flammable.

Important note
Fosroc products are guaranteed against defective materials and manufacture and are sold subject to its standard Conditions for the Supply of Goods and Service. **All Fosroc datasheets are updated on a regular basis. It is the user's responsibility to obtain the recent version.**

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